

ETHAN MERCER

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Principal Game AI & Networked Gameplay Engineer | C++ | Unreal Engine | Large-Scale Multiplayer | Bots | Behavior Trees | Autonomous Simulation

Professional Summary

Fictional senior game developer with 20+ years of experience delivering AAA multiplayer games, Unreal Engine systems, C++ gameplay architecture, large-world simulation, game networking, bots, behavior trees, tactical decision systems, and autonomous-vehicle simulation environments. Career includes fictional leadership roles with globally recognized companies such as Epic Games, Electronic Arts, Riot Games, Ubisoft, and Waymo Simulation Labs. Experience is focused strictly on game development, game AI, and real-time C++ simulation - no Python programming experience is claimed, and no non-game AI / machine learning / generative AI background is included.

Core Expertise

Languages & Engines	Modern C++17/20, Unreal Engine 4/5, Blueprint integration, gameplay frameworks, engine modules, memory/performance profiling, multithreaded gameplay systems.
Game AI & Bots	Behavior Trees, EQS, GOAP-style planning, navigation meshes, tactical cover selection, squad coordination, bot perception, utility scoring, combat directors, large-scale NPC simulation.
Networking	Client/server gameplay, replication, prediction, rollback-friendly systems, dedicated server optimization, matchmaking support, telemetry-driven latency reduction.
Autonomous Simulation	C++ simulation agents, traffic behavior models, scenario playback, sensor mock interfaces, deterministic replay, world-state orchestration for self-driving car test environments.
Explicit Exclusion	No Python programming experience. No data science, machine learning, generative AI, LLM, or non-game AI experience claimed.

Professional Experience

Principal Game AI & Networked Systems Engineer

2021 - Present

Epic Games - Cary, NC / Remote US

- Led C++ and Unreal Engine systems for large-scale multiplayer gameplay prototypes, focusing on authoritative server logic, replication efficiency, and bot-driven test coverage.
- Designed advanced game AI behavior trees for coordinated squads, world traversal, tactical cover, threat evaluation, and role-based combat behaviors in open-world environments.
- Built a deterministic bot harness that simulated thousands of multiplayer sessions per week to expose networking defects, replication edge cases, and server performance regressions.
- Partnered with engine teams to optimize navigation, perception updates, AI tick budgeting, and dedicated-server CPU usage for high-concurrency worlds.

Senior AI Gameplay Architect

2017 - 2021

Electronic Arts - Los Angeles, CA

- Owned C++ architecture for bots, team tactics, movement prediction, animation-aware decisions, and designer-facing Unreal Engine behavior tools across multiple AAA multiplayer titles.
- Introduced hierarchical behavior trees and utility scoring for combat agents, improving designer iteration speed while maintaining deterministic server behavior.
- Implemented network-aware AI decision throttling, bot LOD, and async path request batching to reduce server cost during large matches.
- Mentored engineers on code review, gameplay debugging, memory discipline, Unreal replication patterns, and production-safe feature rollout.

Lead Gameplay AI Engineer

2013 - 2017

Riot Games - Los Angeles, CA

- Developed competitive bot systems for onboarding, practice modes, automated match validation, and large-scale balance testing using C++ server gameplay code.
- Built tactical behavior layers for target selection, lane movement, retreat logic, ability timing, cooldown evaluation, and team objective coordination.
- Collaborated with gameplay designers to convert high-level champion and combat goals into robust behavior tree and state-machine implementations.
- Improved internal simulation stability by adding deterministic replay diagnostics and game-state inspection tools implemented without Python dependencies.

Senior Networked Gameplay Engineer

2009 - 2013

Ubisoft - Montreal, QC / San Francisco, CA

- Delivered C++ multiplayer gameplay features for open-world action titles, including AI population replication, co-op synchronization, and server-validated mission logic.
- Implemented NPC navigation, crowd steering, combat perception, alert propagation, and behavior-tree authoring workflows for dense urban worlds.
- Reduced bandwidth usage through AI state compression, relevancy rules, and carefully staged replication of perception, movement, and combat intent.

Autonomous Simulation Engineer - Real-Time Worlds

2006 - 2009

Waymo Simulation Labs - Mountain View, CA

- Built C++ real-time simulation agents for self-driving car test scenarios, including traffic actors, pedestrian-like agents, route-following vehicles, and reproducible scenario playback.
- Adapted game-style behavior trees and world-state evaluators to model traffic interactions, lane changes, right-of-way negotiation, and scripted edge cases.
- Focused on simulation fidelity, determinism, replay tooling, and high-performance C++ runtime systems; role did not involve Python, machine learning, or model training.

Gameplay Programmer

2002 - 2006

Activision Blizzard - Santa Monica, CA

- Programmed C++ gameplay, enemy behaviors, multiplayer debugging tools, and mission scripting support for console and PC titles.
- Supported dedicated server stability, input prediction, combat encounter logic, and designer-facing debugging displays for AI state and navigation failures.

Selected Project Highlights

- **Massive Bot Validation Platform:** Created C++ Unreal automation that populated multiplayer maps with thousands of behavior-tree agents to stress test matchmaking, replication, pathfinding, and server performance.
- **Open-World AI Director:** Architected systems for spawning, awareness, investigation, pursuit, retreat, and squad coordination across large streaming worlds.
- **Self-Driving Scenario Runtime:** Built deterministic C++ simulation actors and scenario orchestration for autonomous vehicle testing, applying game simulation methods to traffic behavior without machine learning responsibilities.

Education

B.S. Computer Science, Real-Time Interactive Systems

2002

University of Texas at Austin - Austin, TX

Technical Boundaries

This fictional candidate profile intentionally emphasizes C++, Unreal Engine, game networking, bots, behavior trees, game AI, and autonomous simulation. It intentionally excludes Python programming, data science, machine learning, generative AI, LLM systems, web development, mobile app development, DevOps ownership, and unrelated software domains.

Note: This resume represents an imaginary candidate and is not a real employment record.